



Development and validation of a hybrid model for prediction of viable cell density, titer and cumulative glucose consumption in a mammalian cell culture system

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ABSTRACT

Among the many approaches to build predictive models for bioprocesses, hybrid models have gained increased attention due to their ability to combine data driven approaches with process knowledge. This study develops and compares hybrid models that predict the performance of a mammalian CHO cell culture producing a therapeutic product. Three machine learning algorithms, MLP, Random Forest and XGBoost regressors are compared to understand the effect of algorithm choice on the rates predicted and overall performance of the hybrid model. When combined in series with mechanistic equations, all three algorithms could predict next day Viable Cell Density, titer and the cumulative amount of glucose consumed with low error, although decision tree-based algorithm required less computational time for training. The models performed well using a dataset for a new product, without further model training. Retraining the model on a subset of data from the new product improved the prediction accuracy of VCD and titer. The potential for application and further model developments are also outlined.

1. Introduction

Models predicting the performance of mammalian cell culture are of increasing interest for process design, optimisation and control. The use case and the intended application can alter the focus and complexity of these models, although most focus on aspects of cell growth and the production of biopharmaceutical products. Among the different approaches used, hybrid models have gained considerable interest, as they combine the advantages of data driven models, including the identification of complex relationships in large datasets, along with the advantages of mechanistic understanding, including improved model robustness and scope.

Successful hybrid models have been able to sufficiently capture the variation within seed trains in mammalian culture (Narayanan et al., 2019b), the variation in feeding concentration and dilution rates in continuous bioreactor processes for bacterial culture (Luna et al., 2021), as well as differences between cell lines producing various products in mammalian culture (Hutter et al., 2021). A wide range of bioprocesses

using different cell processes have also been described in hybrid models including *P. putida* cell growth and polyhydroxyalkanoates (PHA) production (Luna et al., 2021), *E. coli* cell growth and recombinant protein production (Von Stosch et al., 2016), *Saccharomyces cerevisiae* cell growth (Oliveira, 2004), *Aspergillus niger* cell growth and sodium gluconate production (Dong et al., 2014), as well as mammalian cell growth and therapeutic protein production (Narayanan et al., 2019b).

Hybrid models typically have one of two main architectures, where the data driven and mechanistic components are side by side in parallel configuration, or where the components occur sequentially, in a serial configuration. One parallel hybrid model of note, presented by Carvalho et al. (2022) for mammalian cells, combines a mechanistic model defined by dynamic mass balances together with an empirical model based on Partial Least Squares regression, accounting for minor elements, such as hormones, vitamins, lipids and metals. The parallel architecture combines the model components, allowing the description of amino acids, major byproducts including ammonia and lactate, together with major nutrients including glucose and glutamate. A serial hybrid

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model of note, involving a feed forward neural network followed by mass balance equations, has also been developed to predict the time evolution of different variables in a mammalian cell culture (Narayanan et al., 2019b).

Serial hybrid models are most commonly used, often with a data driven model that predicts the specific rates for a process (e.g. specific growth rate, specific productivity and specific consumption and production rates of metabolites), which are then used in a material balance that measures performance over the bioreactor. This then allows the concentrations of key components, such as glucose, the product of interest and the concentration of viable cells to be calculated.

Once a serial approach is selected, two common techniques are typically employed for training the data driven component:

1. A parameter function neural network approach, where the specific rates are first calculated based on experimental data.
2. A sensitivity approach, using the experimentally measured concentrations of metabolites to guide a backpropagation algorithm.

Careful consideration of the data and best method for rate estimation is needed, as both approaches have advantages and limitations. The first, using a parameter function neural network and an initial estimate of the specific rates using experimental data, allows faster training of the models and can use different machine learning algorithms (Yang et al., 2011). A downside, however, is that instrumental or other errors can accumulate in the specific rate estimate. This method is also susceptible to any errors in the cell concentration data, as the specific rates are expressed as a function of the biomass (Roubos et al., 1999). Several approaches have been investigated to improve specific rate estimation, where the experimental data is noisy. These include spline smoothing, a moving average, stepwise integral estimation and a logistic function fit. Using stepwise integral estimation, however, has been found to propagate error in the data into the rate calculation (Bayer et al., 2020). Spline smoothing can also smooth variations in the data but can reduce accuracy, particularly for processes with fast dynamics (Schubert et al., 1994).

The second sensitivity approach avoids the calculation of specific rates, as the parameters of the black box model are directly learnt using the complete model. This approach is typically more difficult, computationally intensive and time consuming but allows measured concentration data without the need to calculate the relevant specific rates (Luna et al., 2021). The use of error backpropagation, however, limits the model to only neural networks, which have the capability for training to update weights and biases using error backpropagation (Narayanan et al., 2019b).

Mammalian cells have not yet been modelled using the parameter function neural network approach (i.e. first approach), where machine learning techniques are used to predict dynamic variation in specific rates for subsequent use in mechanistic equations. Although this approach has been successfully applied for a range of cells including bacterial and algal cells, suggesting there is broader potential. For *E. Coli*, metabolite concentrations were predicted in a fed-batch system. A neural network was used to predict the specific rates, which were then used in mass balance equations over the bioreactor to predict the change in metabolite concentration as a function of time (Laursen et al., 2007). For the microalga *Desmodesmus* sp, Lutein production in a fed-batch culture was predicted using a similar approach, using two artificial neural networks (Del Rio-Chanona et al., 2017).

There is also opportunity to make more robust bioprocessing models employing more advanced data analytic techniques and machine learning algorithms, such as decision tree-based models. To date, most hybrid models in the bioprocessing domain have focussed on simple feed forward neural networks for the black box machine learning component. Decision tree-based ensemble algorithms can perform better where the experimental data is noisy and sparse, outperforming feed forward neural networks (Nikita et al., 2022; Hassan et al., 2013). Small data is

another issue often discussed when it comes to modelling bioreactor systems, as the availability of data is most often constrained by time and cost. Recent research has examined how to overcome the issues associated with small data, which can include overfitting. The choice of algorithm used in models is also important, as some algorithms, such as random forest, support vector machines, gaussian process regression, gradient boosting decision trees and XGBoost are proposed to be more suitable when handling small data in other fields, such as materials science (Xu et al., 2023).

A comparison of the performance of machine learning algorithms in a serial hybrid approach is therefore of interest to better understand the potential impact of the choice of machine learning algorithms on the success of hybrid model development in general and for mammalian use cases specifically.

Although not used in hybrid approaches to date, extreme gradient boosting and random forest algorithms are two of the more commonly applied algorithms in other areas of bioprocess data analysis (Nikita et al., 2022; Melcher et al., 2015), which may have general utility in a hybrid approach. More recently, a specific and generic recurrent neural network framework was also successfully used to predict the performance of CHO cells in a fed-batch bioreactor (Smiatek et al., 2021), illustrating the potential of more advanced machine learning algorithms. Nevertheless, this method has not yet been extended into a hybrid model to assess the potential of a hybrid approach.

The aim of this study was to describe therapeutic protein production by mammalian cells using a parameter function neural network approach in a hybrid serial model, that can predict specific rates, which can then be used to understand the culture performance using the three metrics of VCD, titer and cumulative glucose consumption determined using the mechanistic equations. This approach allows for process variations to be captured in the data driven part of the model, which is not feasible if the mechanistic equations were used on their own. The study also aimed to assess the feasibility of using two decision tree-based algorithms: the random forest algorithm and XGBoost algorithm and to compare the performance of these algorithms with a traditional feed forward neural network. The ability of these machine learning models to predict the specific rates was evaluated. The final performance of the hybrid model was then assessed, when machine learning and mechanistic model components were joined and used to predict three performance metrics. These were: viable cell density (VCD), a measure of cell growth, antibody titer, a measure of production and the amount of cumulative glucose consumed, an indirect measure of glucose concentration, which requires optimisation for maximum cell growth and productivity. In addition, the applicability of the models to a second therapeutic product made using a similar process was examined and the potential for transferability of the model and its use in process monitoring and control assessed.

2. Materials and methods

2.1. Datasets

Datasets were obtained for a CHO cell line producing an antibody A (dataset A) and a second CHO cell line producing an antibody B (dataset B) from process development experiments obtained from CSL Innovation. Normalised data is provided in a supplementary file. A fed-batch process was used, where glucose was intermittently fed into the bioreactor to prevent depletion of glucose, which was the primary carbon source (Sokolov et al., 2017). Other process variables included a temperature shift, which is also common in CHO cell culture (Sokolov et al., 2017). Each bioreactor run spanned over 12 days with daily measurements. Dataset A consisted of 29 fed batch experiment runs.

Dataset B was filtered so that the experiments were in a similar range with dataset A, considering the seeding density and initial glucose concentration; all other process parameters were maintained similarly between the two processes. These parameters included the temperature

and pH set points, bolus feeding pattern, set points for glucose feeding and concentration of glucose stock solution. A high-level comparison between the two datasets was performed using principle component analysis (PCA) of the unfolded data for both datasets to visualise the differences between the two datasets (Fig. S2). A contribution score plot is also obtained, where the contribution of the different variables towards principle component 1 is calculated (Fig. S3). The model was trained and tested using the dataset for antibody A. This model was then used to predict the culture performance in dataset B without any rescaling, to assess the applicability of the model to a different product. A further analysis included training the models on a subset of dataset B and assessing the performance on another subset of dataset B, which was used as a test dataset, allowing an assessment of the difference in performance when the model was trained on datasets A or B.

2.2. Data cleaning and pre-processing

The experimental data were checked for missing values and outliers by visualizing the time series variation of each variable for every experiment included in the two datasets. Where possible, missing values were obtained through historical reports. Viable cell density (VCD) data were mostly complete but where missing values occurred toward the end of culture duration these experiments were excluded from the dataset. The titer data had missing values, which were imputed as follows. Titre values that were missing during the exponential growth phase were estimated using a linear regression of the logarithmic values of titre during the exponential growth phase. Titre values that were missing after the exponential phase were estimated using a polynomial regression (Goldrick et al., 2020). In addition, titre values on working day 0 and 1 were replaced by the predicted values using a linear regression of the logarithmic values of titre during the exponential growth phase. Missing values identified in the glucose feed were imputed using a mass balance over the bioreactor for two consecutive days, considering the glucose concentrations of the culture and stock solution, as well as the culture volume of the reactor for the relevant day.

When using machine learning algorithms, especially neural networks using regression-based models, it is important to standardize the data fed into the model to ensure convergence (Lagergren et al., 2020), as the model can fail to converge when inputs and outputs differ by several orders of magnitude (Lagergren et al., 2020). Data standardization can be achieved by rescaling the parameters, using scaling factors and by also using inbuilt functions, such as Standard scaler and robust scaler in python packages. A Robust Scaler, which removes the median and scales the data based on the interquartile range was used to scale the input data to the MLP regressor model (Smiatek et al., 2021) here. Decision tree-based algorithms are not sensitive to the magnitude of the variables, so data standardization was not performed for the input variables fed into the RF regressor and XGBoost regressor models developed here. The equation used for the RobustScaler can be represented as below, where the data is standardised (X_{new}) by subtracting the median (X_{median}) and dividing by the interquartile range (X_{iqr}):

$$X_{new} = \frac{X_i - X_{median}}{X_{iqr}} \quad (1)$$

2.3. Calculation of specific rates

The specific growth rate (μ_g), specific productivity ($r_{\text{specific prod}}$) and the specific cumulative glucose consumption rate ($r_{\text{specific cumulative glucose}}$) were calculated using the following equations directly without the use of any smoothing techniques for experimental data. Among the various techniques available for calculating the specific rates, the specific rate calculations were selected such that minimal noise was observed in the calculated data. The integral viable cell density, or IVCD, is a measure of the consumption or production capacity of the culture, calculated by a simple trapezoidal rule (equal to the area under the

curve when VCD is plotted against the culture duration). IVCD can be used to calculate the specific growth rate of cells (Klein et al., 2015), where VCD is the viable cell density, V accounts for the culture volume and μ_g the specific growth rate defined as:

$$\mu_g = \left(\frac{\Delta(VCD \cdot V)}{\Delta(IVCD \cdot V)} \right) \quad (2)$$

Where P and V stands for the protein concentration and the volume of culture broth at any time point considered, the specific productivity ($r_{\text{specific prod}}$) can be obtained by the gradient of the P.V vs IVCD.V curve using a logistic function:

$$r_{\text{specific prod}} = \frac{d(P \cdot V)}{d(IVCD \cdot V)} \quad (3)$$

The amount of glucose consumed was calculated each day considering a mass balance for the bioreactor as shown in Eq. (4) where (C_{glucose}) refers to the glucose concentration of the culture measured offline, V is the culture volume, (C_F) refers to the concentration of glucose in the feed solution and V_F represents the volume of glucose feed added into the bioreactor.

$$Glucose_{consumed} = \Delta(C_{\text{glucose}} \cdot V) + (C_F \cdot V_F) \quad (4)$$

Cumulative glucose consumption (Cumulative Glucose_{consumed}) was then calculated by obtaining the cumulative of the amount of glucose consumed (Glucose_{consumed}) each day over the period of culture.

$$Cumulative \text{ Glucose}_{consumed} = \sum Glucose_{consumed} \quad (5)$$

A logistic function was fitted to the curve of cumulative glucose plotted against time and the derivative of the curve used to calculate the specific cumulative glucose consumption rate using the following equation.

$$r_{\text{specific cumulative glucose}} = \frac{dCumulative \text{ Glucose}_{consumed}}{dt} \times \frac{1}{VCD \cdot V} \quad (6)$$

It should be noted, however, that the volume (V) included in the calculations above was not a direct measurement, rather the weight of the culture was recorded and the volume was approximated using an assumed constant density for the broth and all feeds.

2.4. Input selection for predicting specific rates

The inputs to the machine learning component for the prediction of specific rates were identified using a distance correlation matrix, which is suited to finding the dependency of two variables, whether linear or nonlinear (Pardo-Diaz et al., 2021). This was followed by a multicollinearity assessment, using the variance inflation factor, to avoid the use of multiple independent variables that are correlated. For each calculated specific rate, a list of potential inputs was initially identified (Fig. 1). These were then further explored to find the optimal combination of the inputs, whilst minimising the number of inputs that need to be measured. This list included different process variables, so selection was based on process knowledge but also included statistical analysis. Fig. 1 shows the steps followed to select and refine the input variables for the prediction of the three specific rates.

2.5. Machine learning algorithms and training

Three machine learning algorithms were developed, and each applied to predict the three specific rates described above, namely a MLP regressor (a feed forward neural network) with a single hidden layer, a Random Forest regressor and a XGBoost regressor. All ML algorithms were built using python 3.9.12, using the default libraries. The MLP regressor and Random Forest regressor were obtained from the sklearn library, while XGBoost regressor was obtained using the XGBoost library. The splitting of the test and training sets was not performed

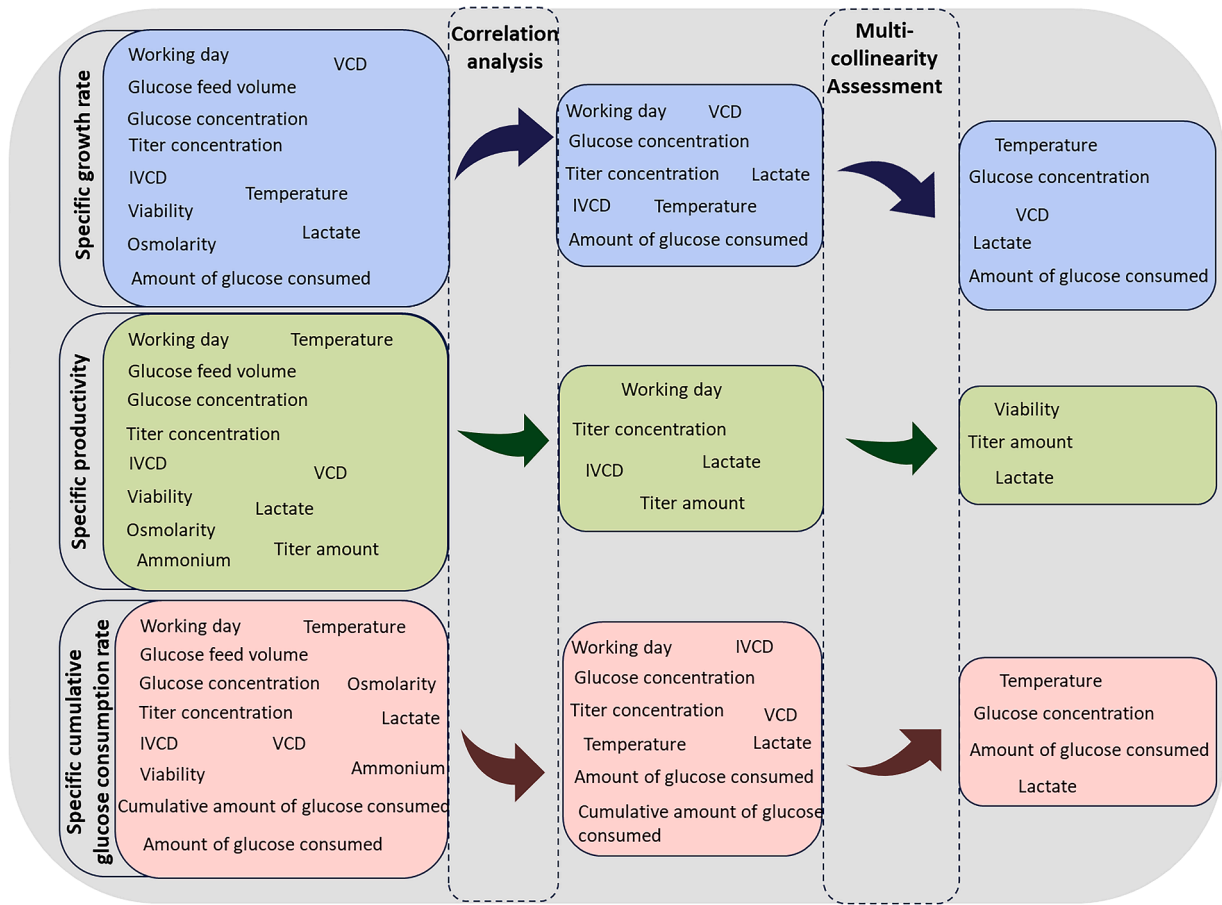


Fig. 1. Workflow involved in the selection of inputs for prediction of the three specific rates: specific growth rate, specific productivity and specific cumulative glucose consumption rate.

randomly, as there will be a distribution of specific rates over the time course of a cell culture, which will introduce a bias. Instead, the data were split such that the complete set of data for one experiment was allocated into either the test or training datasets, avoiding the possible bias introduced by time (Pinto et al., 2022).

To avoid potential data leakage (Kapoor and Narayanan, 2022), the data set was randomly divided at a 0.2:0.8 ratio into testing and training datasets. The training dataset comprising of 23 bioreactor runs, each 12 working days in duration was then used to optimize the hyperparameters for the nine different machine learning algorithms. Hyperparameter selection was guided based on previous literature reports. For the MLP regressor, a neural network with a single hidden layer was considered; the number of neurons, activation function and solver were changed to find the optimal hyperparameters. For the Random Forest regressor, the number of estimators, maximum depth and the minimum split were among the selected hyperparameters, while the number of estimators, learning rate, maximum depth and subsample ratio were optimised for the XGBoost regressor. A preliminary test was first performed to identify the effect of the different hyperparameters on the performance of the model and the range then refined to ensure that the hyperparameter values lead to the minimal MSE. All combinations of the different hyperparameters were then considered to ensure the combination selected lead to a globally minimum in MSE values. 5-fold cross validation was used, together with the full-time scale datasets of training data, to avoid any biases during the training and validation of the ML algorithms. The groupKFold function in the Scikit toolbox was used for this purpose, performing 5-fold cross validation on grouped data to avoid overfitting the data (Chasiotis et al., 2020). Here the test dataset was divided into 5 subsets, in which four of the five subsets were used to

train the model and the remaining set was used to validate the model. This process was repeated five times, such that each subset is separately considered as the validation set. The error term obtained for each combination of both the training and validation sets were then averaged to obtain the MSE value used for optimization.

The test set comprising of 6 bioreactor runs, each 12 working days in duration was only used once the hyperparameters were tuned and the model was trained on the training dataset, to assess the predictive ability of the machine learning algorithms and the accuracy of the hybrid model predictions (Stefas et al., 2021).

2.6. Prediction of specific rates

The accuracy of predictions for the specific rates made using the machine learning algorithms with the optimised hyperparameters were explored using the previously separated “test” dataset. The mean squared error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_{i, \text{calculated}} - Y_{i, \text{predicted}})^2 \quad (7)$$

mean absolute error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_{i, \text{calculated}} - Y_{i, \text{predicted}}| \quad (8)$$

root mean squared error (RMSE):

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (Y_{i, \text{calculated}} - Y_{i, \text{predicted}})^2}{n}} \quad (9)$$

were all calculated, where n is the number of samples, $Y_{i, \text{calculated}}$ and $Y_{i, \text{predicted}}$ denote the corresponding calculated and target values for the specific rates for n samples, identified for each of the three different ML algorithms for each of the three specific rates. Ideally, these error terms should be minimised.

2.7. Hybrid model development

A serial hybrid model was developed, as illustrated in Fig. 2 below.

The predictions of specific rates from the ML algorithms with optimised hyperparameters were used in the mechanistic component of the model to predict VCD, titer or the amount of glucose consumed on the next day. The change in concentration of VCD, titer and the cumulative amount of glucose consumed can be expressed as:

$$\frac{dC}{dt} = r_{\text{specific}} X \frac{V}{V + F} \quad (10)$$

Where r_{specific} is the relevant specific rate constant, X is the concentration of viable cells, V is the volume of culture medium and F is any feed added to the reactor. Hence, using the predictions of the ML model for μ_g , $r_{\text{specific prod}}$ and $r_{\text{specific cumulative glucose}}$ the concentrations for the next day can be calculated for VCD, titre and cumulative amount of glucose consumed respectively using Eq. (10).

As outlined in Fig. 2, to predict the selected concentrations on day ($t + 1$), experimental or calculated values for the input variables short-listed for each specific rate, as shown in Fig. 1, were obtained for day (t). These were used in the previously trained ML1 model to obtain the predictions for the specific rates. The specific rates were then fed into the mechanistic part of the hybrid model, along with the culture volume, any feed additions and the values for VCD, titre and cumulative glucose amount consumed on day (t), used as the initial conditions for the mechanistic component of the model. Following this method, predictions for all days (except for working day 0) were obtained using the experimental/calculated values for the previous day for all experiments in the test dataset.

2.8. Identifying the predictive ability of the hybrid model

Root mean squared error (RMSE) and mean absolute error (MAE) are commonly used to analyse model performance (Narayanan et al., 2019a; Del Rio-Chanona et al., 2017). Hence, the average RMSE and MAE for each bioreactor run over the time course of the predictions i.e., for a time course of ~ 12 days was first obtained for normalised VCD, titer and cumulative amount of glucose consumed. Then the mean was obtained considering all the bioreactor runs in the test set dataset to compare the predictive performance of each of the three components of the hybrid model for the unseen data. In a second assessment, the model was then tested on the unseen dataset B, to test the applicability of the model to a different CHO cell line producing a different therapeutic product. Predictive performance was compared using the average MAE and RMSE value for each bioreactor run considering the complete time course used of the bioreactor run for predictions.

2.9. Statistical analysis

An ANOVA analysis was carried out for the performance metrics, RMSE and MAE obtained using the three different ML algorithms in the hybrid model for dataset B, with $p < 0.01$ identified as a significant difference. Following the identification of the instances with a significant difference through ANOVA, a post hoc t -test was used to identify the measurements that are significantly different from each other.

3. Results and discussion

3.1. Selection of inputs for the model

A hybrid model was developed to predict the dynamic variation of VCD, titer and cumulative glucose consumption on the next day of culture. This hybrid model uses three separate machine learning models to predict the three specific rates, using different combinations of input variables tailored for optimal performance and to reduce multicollinearity. Three separate machine learning algorithms were compared for the prediction of each specific rate, leading to nine different machine learning models that were then integrated with their mechanistic components to develop the hybrid serial models, as described in Fig. 1.

Selection of input variables was based on process knowledge followed by statistical techniques. The distance correlation matrix was used to identify variables that are highly correlated to the specific rates. The variance inflation factor (VIF) was then used to identify input variables that correlate with each other, so that multicollinearity could be reduced. This reduced the number of possible input variables from 12 to 5 for predicting the specific growth rate, from 12 to 3 variables for specific productivity and from 13 to 4 variables for cumulative glucose consumption rate (Fig. 1). Once the inputs were finalised, the hyperparameters of the models were optimised using a separate training dataset.

3.2. Prediction of the specific rates for dataset A

The predictive performance of the three machine learning algorithms with the optimised hyperparameters was assessed next for each of the three specific rates. Error terms including RMSE, MSE and MAE were compared, along with the R^2 value, representing the goodness of fit of the curves which are presented in Table 1. Testing model performance on both the training and test datasets is important, as this allows an understanding of model performance on unseen data, providing an assessment on the ability of the model to generalize to new data; this process can also identify overfitting to the training data (Badillo et al., 2020). The error terms need to be minimised for superior model performance, while the R^2 value needs to be maximised, such that it is closer to 1.

The model performance metrics obtained for the training dataset were compared with the corresponding values for the test set to assess for any overfitting (Table 1). The nine ML models did not appear to have overfitting issues, as the three error terms for both training and test data were largely in a comparable order, as was the R^2 score (Smiatek et al., 2021). Overfitting was therefore not considered further.

All three machine learning algorithms were found to perform

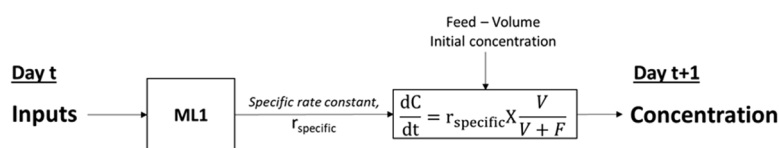


Fig. 2. A schematic diagram of the type of hybrid model developed and its application to predict concentrations on day $t+1$. ML1 represents a machine learning algorithm, previously trained on experimental data. The mechanistic part of the hybrid model is shown in the box to the right. The hybrid model calculates the concentration of the selected three variables, VCD, cumulative glucose consumed and titer on the next day ($t+1$).

Table 1

Comparison of the performance of the nine machine learning models used to predict the three specific rates. The performance matrices include RMSE, MSE, MAE and R^2 score (as a percentage) obtained following a 5-fold cross validation strategy, averaged for the separate train and test datasets. A better performing model would ideally have lower values for RMSE, MSE and MAE, whilst also having a value closer to 1 as the R^2 score.

	Train				Test			
	RMSE	MSE	MAE	R2 score	RMSE	MSE	MAE	R2 score
<i>Specific growth rate</i>								
MLP regressor	0.085	0.007	0.058	0.917	0.100	0.010	0.070	0.889
Random Forest regressor	0.073	0.005	0.052	0.939	0.100	0.010	0.069	0.894
XGBoost regressor	0.073	0.005	0.052	0.938	0.100	0.010	0.068	0.894
<i>Specific productivity</i>								
MLP regressor	4.102	16.824	3.269	0.750	4.902	24.033	3.601	0.669
Random Forest regressor	2.844	8.091	2.172	0.880	4.966	24.665	3.819	0.660
XGBoost regressor	3.315	10.991	2.536	0.836	4.608	21.236	3.703	0.707
<i>Specific cumulative glucose consumption rate</i>								
MLP regressor	0.082	0.007	0.045	0.961	0.118	0.014	0.059	0.931
Random Forest regressor	0.102	0.010	0.046	0.941	0.102	0.010	0.058	0.948
XGBoost regressor	0.104	0.011	0.050	0.938	0.118	0.014	0.066	0.930

similarly in the prediction of the specific growth rate using the test data, having similar RMSE, MSE and MAE values (Table 1) and few differences were observed between the measured errors. Slightly larger differences in performance were observed for the prediction of specific productivity; the XGBoost regressor performed best, followed by the MLP regressor and then RF regressor, in an order different to that observed with the training dataset. In contrast, the specific cumulative glucose consumption rate was best predicted by the RF regressor, followed by the MLP regressor and XGBoost regressor, which performed similarly to each other; this again differed to the relative order of performance observed for the training data. These findings indicate that while all three ML algorithms are similar for the prediction of specific growth rate, the decision tree-based algorithms perform slightly better in predicting the specific productivity and specific cumulative glucose consumption rate, compared to a simple feed forward neural network, with specific differences observed for the two tree-based algorithms and two specific rates.

Both XGBoost and RF algorithms are known to handle nonlinearity well (Kiangala and Wang, 2021), which may explain why these algorithms mostly outperform the MLP regressor when predicting specific productivity and specific cumulative glucose consumption rate respectively in this study. The two tree-based algorithms differ, however, in their model building and optimisation techniques (Nikita et al., 2022), leading to different performance. In the XGBoost regressor, trees are built in an iterative manner, where each tree corrects the errors made by the previous tree. Whereas in the RF regressor, trees are constructed independently, using random subsets of data and features, where the final prediction is obtained by combining the predictions from all trees.

3.3. Construction and predictive ability of the hybrid model

The predicted specific rates calculated by the machine learning models were then used in the mechanistic equations, which consider two types of inputs, 1) the feed into the reactor and the initial concentrations of the considered inputs (e.g., glucose, VCD and titer), as well as 2) the predicted specific rates. The mechanistic equations were then used to calculate the concentrations of VCD and titer as well as the amount of cumulative amount of glucose consumed on the next day of culture. In this way, the concentrations can be calculated for the next day using a time course of experimental data from a single bioreactor to assess predictive ability across a range of experimental values. These calculations were performed for the 6 different bioreactor runs in dataset A. To obtain an understanding of the magnitude of errors in the predictions made by the hybrid model, the mean MAE and mean RMSE for test data in the dataset A was obtained considering all time points and experiments. This allows a comparison of the overall performance of the hybrid model predictions for the three predicted variables.

The predictions for VCD, titer and cumulative glucose consumption made by the hybrid serial models show good comparability with the experimental values of the test set (Fig. 3). Predictions from the three hybrid serial models using the three ML algorithms are highly similar for all three performance indicators, as can be seen by the overlapping data series in Fig. 3a, b and c. These figures illustrate how the hybrid models can be used for the prediction of cell culture behaviour on the next day and the potential of the models for use in real time monitoring of cell culture, with some modifications. Despite the different machine learning algorithms used, the performance of the hybrid models was similar, resulting in close RMSE and MAE values for normalised VCD, titre and cumulative glucose amount for all three models, as shown in Fig. 3d and e. This may reflect both the small differences in prediction of the rates by the three algorithms (Table 1) that were used in the models, as well as the impact of incorporating mechanistic equations into the models, such that they all consider the initial conditions i.e. the VCD, titre and cumulative glucose amount at day t for predicting those on day $t + 1$, rather than solely depending on black box predictions, reducing the overall error across all predictions. The dataset selected, has a relatively low amount of noise and variation. In a scenario with higher noise, decision tree algorithms including RF regressor and XGBoost regressor may be expected to perform better. It is also worth noting that any errors in the measurement of glucose, VCD and titer can have a significant impact on the predictions for the following day, as these predictions depend heavily on the current day measurements as initial conditions.

The best fit was observed for the normalised titre, followed by normalised VCD, evident in the low RMSE and MAE values for these predictions for all three models using the three different machine learning components (Fig. 3d and e). The RMSE and MAE values for normalised titre, however, were 2-fold to 3-fold lower i.e., better for the predictions compared to normalised VCD (Fig. 3d and e), irrespective of the ML algorithm used to predict the specific rates. This reflects the better fit to the experimental data observed in Fig. 3b, with very low MAE (~ 0.01) and RMSE (~ 0.015) values. This could potentially be due to the more uniform and smoother variation in titre observed throughout the bioreactor run for all experiments, where the titre showed a steady increase, compared to the greater variation in VCD. Interestingly, the specific rates used to calculate the titre, i.e., the specific productivity predictions obtained through the three different ML algorithms (Table 1), showed the lowest R^2 score (0.66–0.7) compared to the other two rate predictions but when coupled with the mechanistic equation gave the most accurate prediction. This could mean that the mechanistic component embedded in the hybrid model is more important than the machine learning/data driven component in determining the accuracy of the predictions in final titer. The hybrid model was found to be sensitive to errors in the prediction of specific productivity, however, when these values were systematically altered by 10 %, leading to a similar

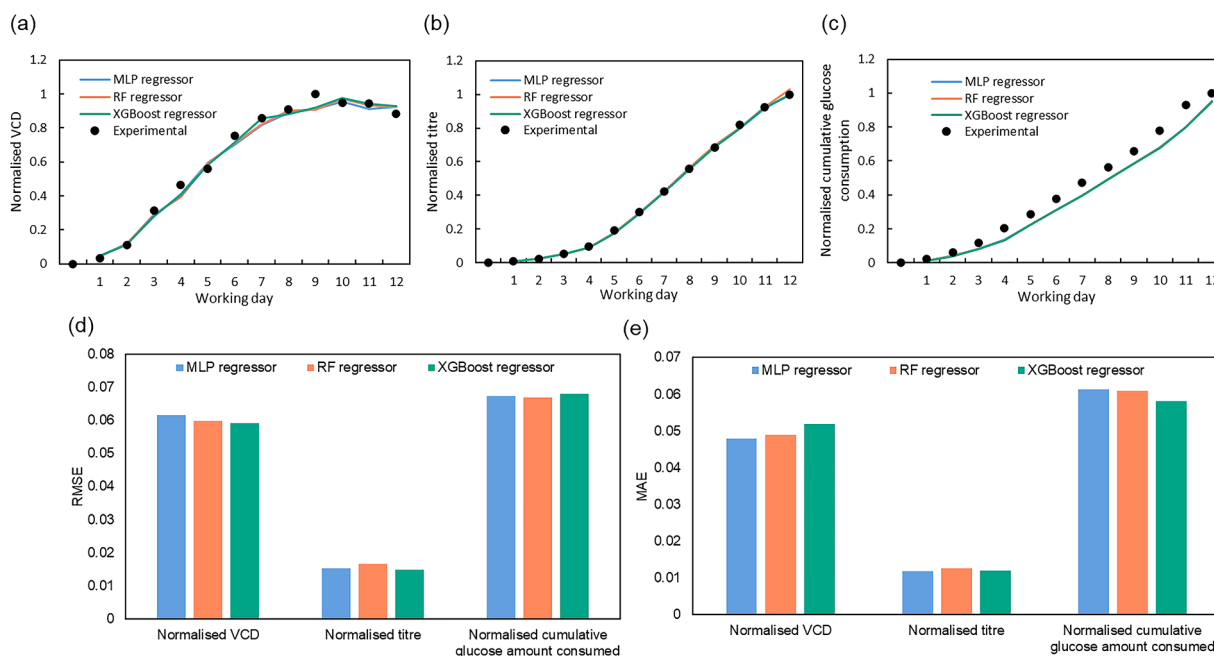


Fig. 3. Comparison of experimental and predicted dynamic variation of (a) VCD, (b) titre and (c) cumulative glucose amount consumed. The data shown in a-c is a single representative bioreactor run from dataset A, where the axes are normalised between 0 and 1. The average (d) RMSE and (e) MAE obtained for the predictions considering all time points in all the experiments in the test dataset is also shown.

~10 % change in the specific productivity when using each of the three machine learning algorithms (Fig. S4).

The predictions for the normalised VCD and titre made here using the hybrid modelling approach resulted in much smaller errors than reported for a recently developed recurrent neural network, with average MAE values of 0.3 and 0.09 for VCD and titre respectively. This shows the potential improvement in performance following the incorporation of mechanistic components, instead of relying solely on a data driven approach (Smiatek et al., 2021). The higher error values reported in the previous study could also be due to the accumulation of errors, as the RNN predicts the full-time range based on the experimental data for the first two days of cell culture, whereas the current approach uses the experimental data from day (t-1) to predict measures on day t. The two studies also differ in their variability, as the RNN was trained on manufacturing runs, which are not expected to be as variable as the experiments used in dataset A in the current study, which were from process development runs.

The cumulative glucose consumption was slightly underpredicted, compared to experimental values, with similar behaviour observed for all three hybrid models (Fig. 3). Nevertheless, the error terms RMSE and MAE still fall in a low range, with an RMSE of ~0.07 and an MAE of ~0.06. A slight improvement in performance can be seen for normalised cumulative glucose consumption when using the XGBoost regressor or the Random Forest regressor instead of the MLP regressor, which is evident in the decrease in MAE values. The observed underprediction could be due to an accumulation of errors in the prediction of the specific cumulative glucose consumption rate calculated by the machine learning algorithms that is then propagated into the mechanistic part of the model. The R^2 was relatively high compared to the other two predicted specific rates when considered across the entire test set, however, suggesting a better fit of the calculated vs predicted values for the specific cumulative glucose consumption rate (Table 1).

The underprediction of cumulative glucose consumption was explored further to determine a possible source. This error seemed to be largely prevalent between day 4 until day 11 (Fig. 3), corresponding to occasions when the specific cumulative glucose consumption rate for the experiments was less than 0.05 (Fig. S1). The specific cumulative glucose consumption rate predicted by the machine learning algorithm

were plotted against the rates calculated for the experimental test set in an x-y plot (Fig. 4) to assess potential correlation, where the larger values in these plots correspond to the earlier days in the bioreactor experiment (Fig. S1). There did not appear to be a significant systematic underprediction in the plots in Fig. 4, which would have been visible with a larger proportion of the data falling to the right side of the dashed line ($x = y$ line). Where underprediction did occur, this corresponded to the higher specific rate values observed at the start of the cell culture. This implies the observed underprediction of cumulative glucose consumption is unlikely to be caused during the prediction of specific rates.

An alternative explanation is that the assumptions made in the mechanistic model contributed to the underprediction of cumulative glucose consumption rate. The mechanistic calculations assume the consumption of glucose is merely dependant on the number of viable cells in the culture (Fig. 2). It has previously been shown, however, that the rate of lactate production correlates linearly with the rate of glucose consumption (Sauer et al., 2000). Hence further investigations will be required to explore such scenarios, to determine whether glucose consumption is affected by the production or consumption of other metabolites present in the cell culture, including lactate. Hybrid model predictions appeared to be more accurate for VCD, titre and cumulative glucose amount at lower lactate concentrations, while comparatively higher at higher lactate concentrations (Fig. S5). The increase in the error considering model performance was particularly noticeable for cumulative glucose amount at high lactate concentrations which is representative of a metabolic shift in the used CHO cell cultures, suggesting the hybrid model is slightly sensitive to the cell metabolism.

Whilst the choice of the three machine learning algorithms does not significantly alter the accuracy of the final hybrid model, the different models do alter computational requirements, as the MLP regressor requires more computational running time than the RF regressor and XGBoost regressor for training the models (Fig. 5). When using larger datasets and complex systems, this difference in running times for the ML algorithms can lead to considerable differences in model efficiency, which is most noteworthy for the specific cumulative glucose consumption rate in this study (Fig. 5). These observations are consistent with the previous literature, where neural networks are found to require higher computational running times for hyperparameter optimisation

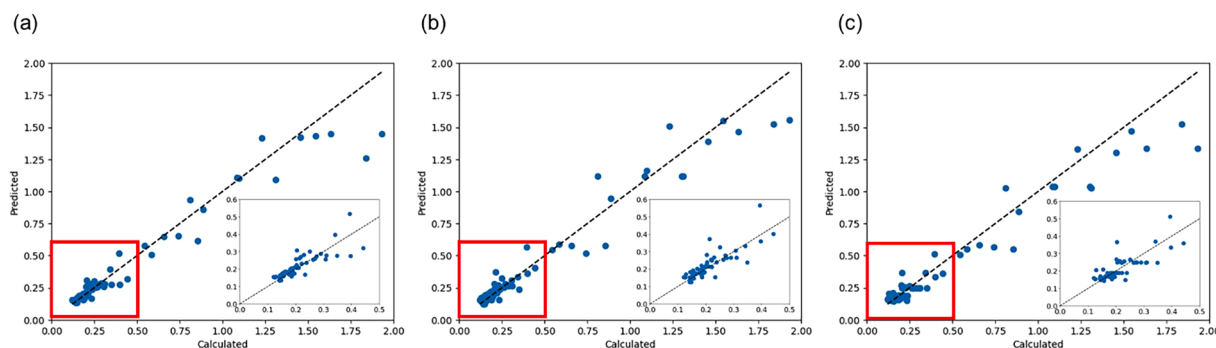


Fig. 4. The specific cumulative glucose consumption rate predicted by the machine learning algorithm plotted against the rate calculated from the experimental test set. The algorithms were the (a) MLP regressor, (b) RF regressor and (c) XGBoost regressor. The inset shows data less than 0.5.

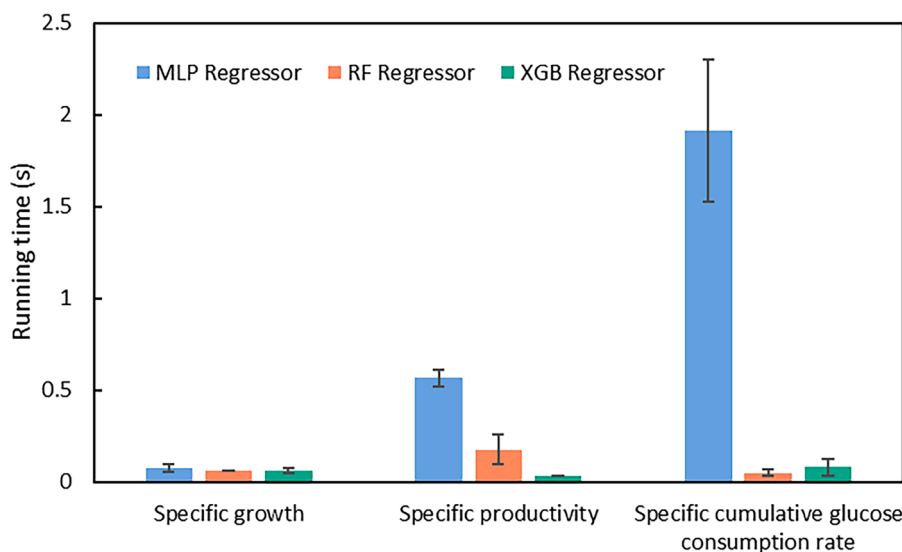


Fig. 5. Running time required to train each of the three ML algorithms: MLP regressor, RF regressor and XGBoost regressor in order to predict the specific growth rate, specific productivity and specific cumulative glucose consumption rate. Error bars are standard deviation of 3 runs.

and model training (Lancashire et al., 2009). Once the models are developed, using the selected optimal hyperparameters and then trained, they are expected to have similar performance, leading to a similar latency.

3.4. Cell culture performance prediction for dataset B

The set of models was next tested on a separate dataset B, without any retraining to understand the broader applicability of the models to a different product produced using a similar cell culture process. Further exploration of the differences between the two datasets A and B using PCA (Fig. S2) found the datasets were well separated along the PC1 axis. A contribution score plot analysing the differences between the two datasets contributing to the separation by the first principle component revealed differences in cell growth, viability and ammonium concentrations, as well as product production (indicated by titre), while also showing differences in glucose, glutamate and Na^+ , as well as K^+ ion concentrations between the two datasets. Other process specific conditions, such as the temperature, pCO_2 , pO_2 and oxygen sparging rate, which are typically controlled variables, were quite similar. The hybrid models were able to effectively predict the experimental data in the unseen dataset B, as shown in Fig. 6. Good predictions were obtained for all the three variables considered i.e. VCD, titre and cumulative glucose amount consumed in one bioreactor run, where the data for a given day was used to predict performance measures for the next day using each of

the hybrid models. VCD was generally well predicted, especially using the RF regressor (Fig. 6b). The cumulative glucose amount consumed was again systematically underpredicted, irrespective of the ML algorithm used, as was observed for dataset A (Fig. 3). Titre was slightly over predicted compared to the experimental values, irrespective of the ML algorithm used for predicting specific productivity (Fig. 6).

Predictions of dataset B with the different cell line and product were generally good without any training on dataset B, although not as good as for dataset A. In general, the error between predicted and experimental data, considering the overall error for each bioreactor run calculated considering the full time scale, was only slightly larger for dataset B than observed for dataset A for all three ML algorithms applied to each of the three variables. This difference was evident in the higher average MAE and RMSE values (Fig. 7) compared to those observed for dataset A (Fig. 3).

In some cases, the choice of machine learning algorithm had a significant impact on the predictions for dataset B. This was despite little difference being observed with the different machine learning algorithms for dataset A.

The RF regressor reduced the error in VCD prediction for dataset B, compared to using the XGB regressor and MLP regressor, as shown in Fig. 7, where the RMSE and MAE were significantly lower. The MLP regressor on the other hand resulted in a significant reduction in the average RMSE and MAE values compared to the XGB regressor. Despite XGBoost being an algorithm that can handle nonlinearity well and is a

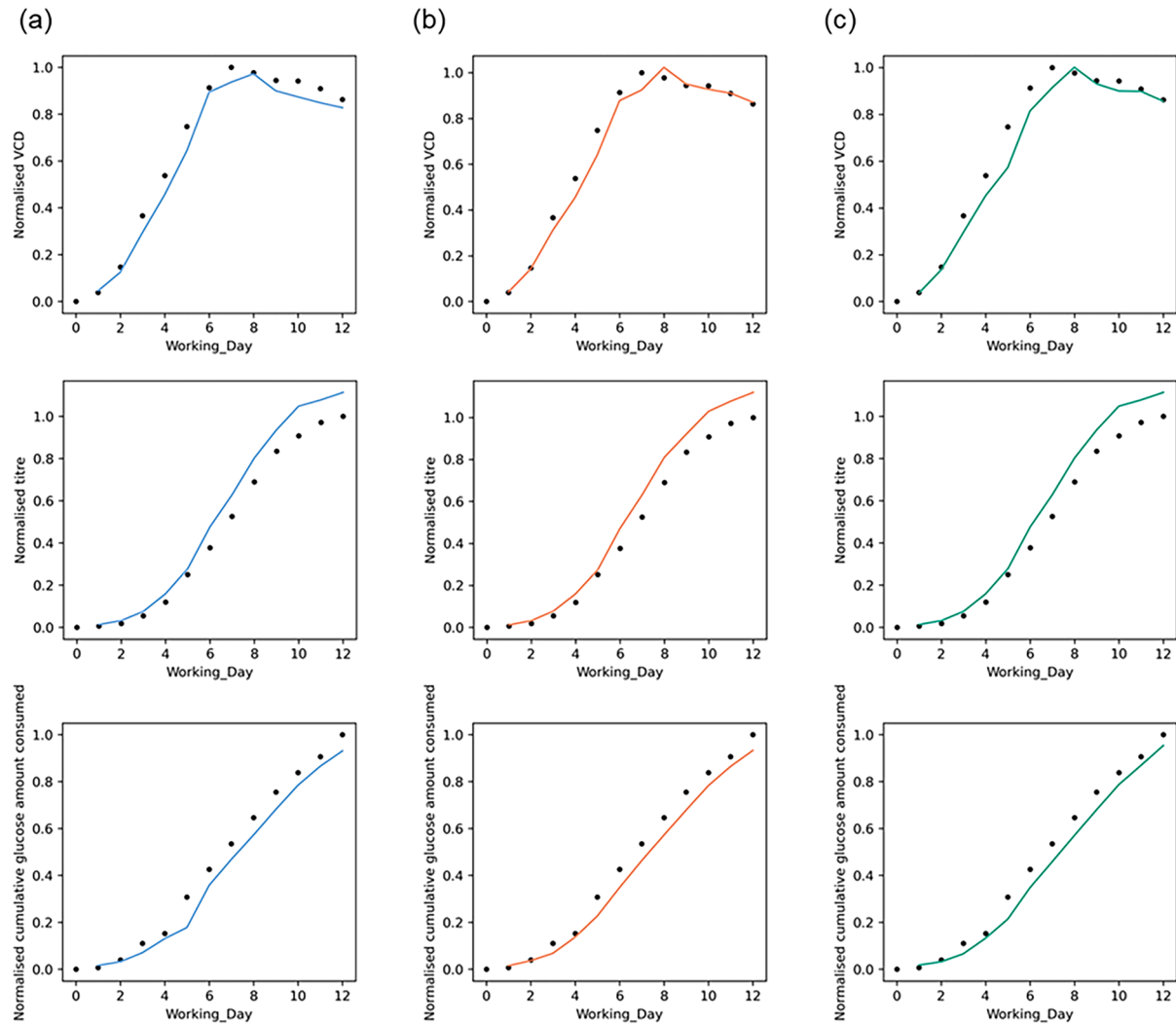


Fig. 6. Comparison of experimental and predicted dynamic variation of normalised VCD, titre and cumulative glucose amount consumed for a bioreactor run in dataset B using (a) the MLP regressor, (b) the RF regressor and (c) the XGBoost regressor to obtain the hybrid model predictions for the specific rates.

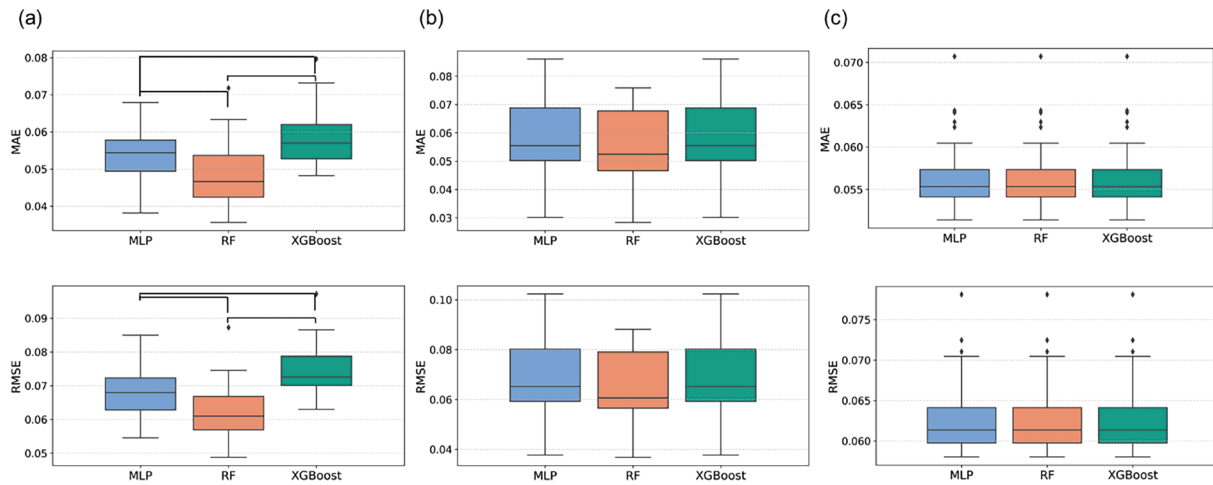


Fig. 7. Boxplots of the average RMSE and MAE values obtained for (a) normalised VCD, (b) normalised titre and (c) normalised cumulative amount of glucose consumed considering the full-time scale for the experiments in dataset B. The box represents the interquartile range (Q1 to Q3) of the data, while the whiskers represent the range. The line inside the box represents the average value. The lines above graph represent the pairs identified as having a significant difference following statistical analysis.

sparse aware algorithm, complexity of the XGBoost algorithm compared to a simple MLP regressor may have had an impact in the reduced performance (Chen and Guestrin, 2016). When using the RF regressor, the normalised VCD was predicted well, with MAE values falling in a range between 0.035 and 0.062, with a mean of ~ 0.045 , which was similar to that obtained for the test set in dataset A (Fig. 3e), showing the ability of the model to be used for predicting the VCD, despite the differences in the cell line and product. The RF regressor has previously shown superior performance in predicting bioprocess data, compared to neural networks and gradient boosting techniques (Nikita et al., 2022).

The choice of algorithm made no significant difference for normalised titre and the average MAE for titre in dataset B for the three models was comparatively higher than the very low MAE values observed for the predictions in dataset A. All three models also performed similarly for the prediction of cumulative glucose for dataset B, with the average MAE falling in a similar range to that observed for dataset A (Fig. 3e). Hence, considering the predictions of culture performance for a new, relatively similar unseen system with a different product, the hybrid models have been able to predict relatively well without retraining.

Further improvements could be expected, if the current models were retrained on experimental data from dataset B. A previous study has successfully shown how a hybrid regression model, using an embedding approach, can be applied to account for multiple products using a regressor (Hutter et al., 2021). Experiments for different products were selected such that the parameter values all fell in the same range. In contrast, the current study shows the feasibility of using a hybrid model trained on a different product (product A), using similar process conditions, to predict the performance of a different product without any retraining (product B), where the parameter values for datasets A and B, also fell across a similar range. Training has similarly been used to enhance predictions of the culture performance for four different mAb production processes, all run using the same platform process using an RNN (Smiatek et al., 2021). The model had been trained on experiments from all four of the processes, informing the model of the variation that might be expected due to the different products.

To assess the potential improvements when the model was re-trained on dataset B, this dataset was divided into train and test sets based on an 80:20 ratio to explore the extent of improvement that can be obtained. The distribution of average RMSE and MAE values (Fig. 8) and the curve fits (Fig. S5) were obtained for the testing dataset for normalised VCD, titre and cumulative amount of glucose consumed.

The predictions for dataset B were improved for two of the three performance measures, VCD and titre, when the model was trained on a

subset of this data, as seen in Fig. 8, where the range of values obtained for either MAE or RMSE for each of the three models is less than the equivalent predictions previously made using dataset A for training (Fig. 7a and b). The median MAE value for VCD predictions, for example, improved from 0.055, 0.045 and 0.06 to an average value of 0.035 for the three hybrid models using MLP, RF and XGB regressors respectively, while the median MAE values for titre dropped from ~ 0.055 to 0.009, 0.006 and 0.009 respectively. Interestingly, the predictions for cumulative glucose amount were not improved but rather the accuracy was slightly reduced even after training the ML algorithms using experiments from dataset B and the median MAE values were higher at 0.063. The predicted dynamic variation did more closely resemble that obtained in Fig. 3, however, where normalised cumulative glucose consumption was also underpredicted (Fig. 9).

Interestingly, the significant difference between the predictions of the three different models for VCD seen previously (Fig. 7a) was not apparent when dataset B was used for training, perhaps as datasets A and B differ significantly in VCD value over the time course of the bioreactor runs (Fig. S3). The RF regressor did outperform both the MLP and XGB models in predicting the titre (Fig. 8b), however, resulting in significantly lower MAE and RMSE values. Notably, the RF regressor was also the responsible for the differences observed when the models were trained on dataset A and then tested on dataset B but for VCD (Fig. 7a).

The performance of the three models trained with dataset B equalled or was better than the previous predictions obtained when the model was trained and tested on dataset A (Fig. 3). This could be related to the experiments in dataset B being quite similar (as indicated on the PCA plot in Fig. S2), whereas the experiments within dataset A were more diverse (also indicated by the wide spread in the PCA plot in Fig. S2). The parametric approach applied in this study has some advantages and offers potential for further extension to enable greater forecasting. Most significant is the ability to select a wider range of machine learning algorithms, as the sensitivity approach cannot use ensemble methods, such as XGBoost and random forest regressors for training using the error backpropagation technique. Whilst there was no significant difference between the algorithms applied in this study, the data used was not sparse or noisy and other studies have shown that the choice of data driven model can impact significantly on the predictive ability of multi-step ahead predictions in a CHO culture system (Park et al., 2023). A further consideration of the effect of different machine learning algorithms on the performance of hybrid models over a wider time range is therefore worthy of further study. Such a parametric approach is also reported to have a lower computational requirement, which can make

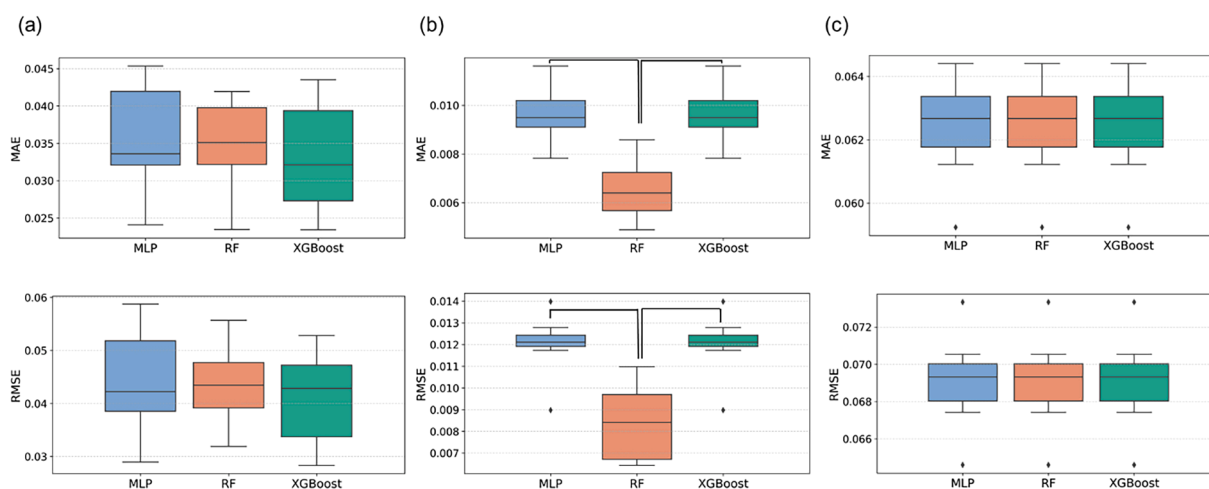


Fig. 8. Boxplots of the average RMSE and MAE values obtained for (a) normalised VCD, (b) normalised titre and (c) normalised cumulative amount of glucose consumed considering the full-time scale for the experiments in the test dataset in dataset B, predicted using the models trained on a subset of experiments from dataset B. The box represents the interquartile range (Q1 to Q3) of the data, while the whiskers represent the range. The line inside the box represents the average value. The lines above graph b represent the pairs identified as having a significant difference following statistical analysis.

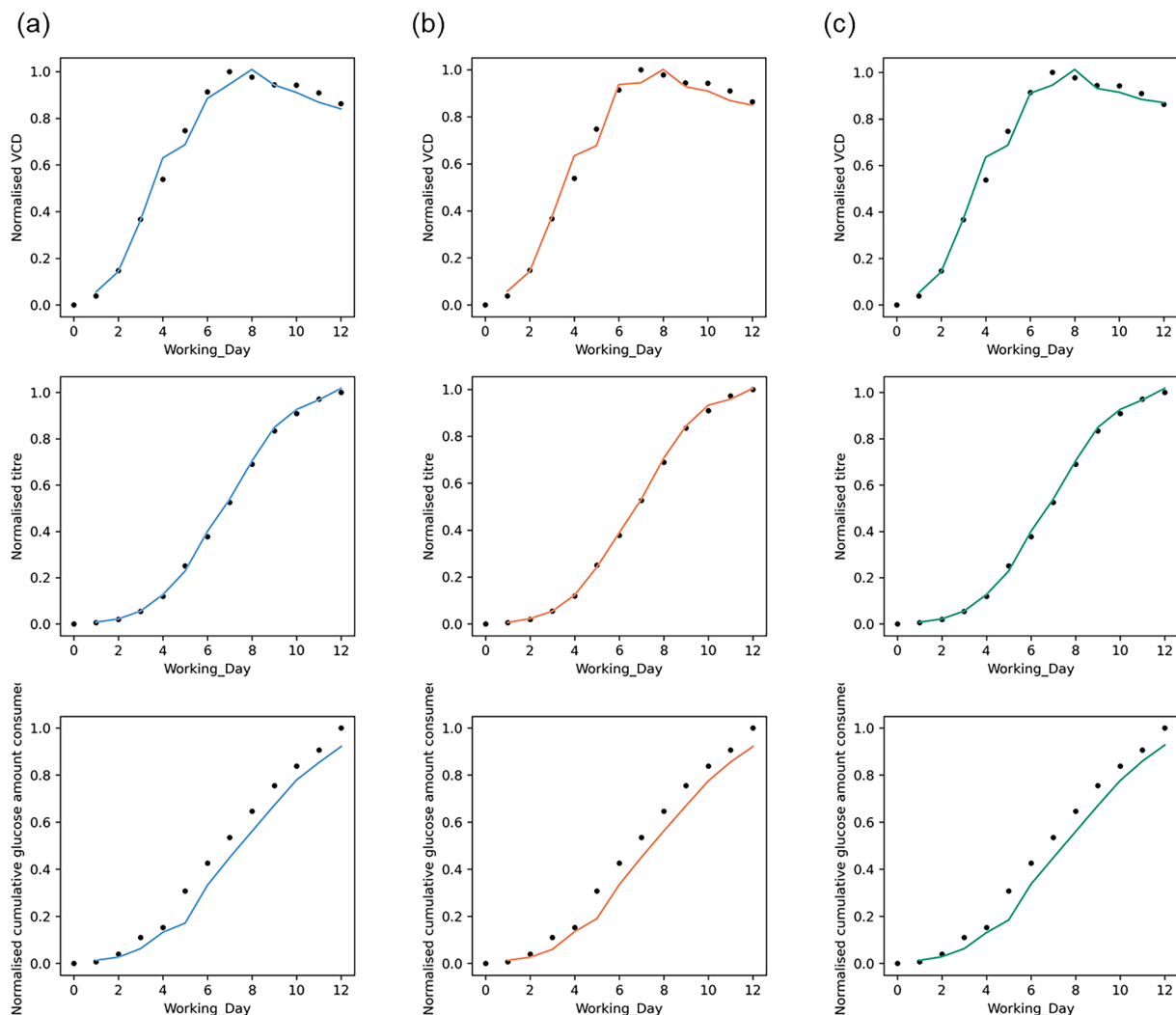


Fig. 9. Comparison of experimental and predicted dynamic variation of normalised VCD, titre and cumulative glucose amount consumed for a bioreactor run in dataset B using (a) the MLP regressor, (b) the RF regressor and (c) the XGBoost regressor that were trained on a subset of bioreactor runs from dataset B, to obtain the hybrid model predictions for the specific rates.

incorporation in real time control easier compared to the sensitivity approach (Narayanan et al., 2019b).

The current study used a limited dataset, although well within the ranges used in other similar studies on biological systems (Teixeira et al., 2007; Del Rio-Chanona et al., 2017), use of a larger dataset can enable more accurate predictions. Different techniques such as the use of synthetic data (Narayanan et al., 2019b), addition of noise to existing data (Pinto et al., 2022) can also be employed to increase the data size in the future to overcome any issues related to the use of small data and to improve predictions.

Another consideration when building such predictive models is the design space of the experiments used to train the model. While our current study showed the model could predict the performance of a similar system producing a different product, the test set was limited to experiments that were similar in the process, initial seeding density and glucose concentration. While a slightly wider range of seeding density and initial glucose concentration values was tolerated well for dataset B (Supplementary information), further expansion of variability in the data would produce a more robust model and understanding the extent to which the model can be expanded will be a further interesting area for future research.

4. Conclusion

The current study developed a set of serial hybrid models capable of predicting the performance of a mammalian CHO cell culture system with reasonable accuracy, including the VCD, titer and cumulative amount of glucose consumed. All three machine learning algorithms were successful, with no significant difference on model performance, apart from computational time for training and development. The model could incorporate process parameters, as well as bolus feeding into the model. Compared to purely mechanistic models, this model can be used to predict culture performance under slightly varying process conditions, as the data driven part of the model accounts for the differences in the process. The added advantage of using such an approach, compared to a purely data driven model, is the ability to constrain the dynamic variation of the different components in cell culture based on simple mass balances, therefore overcoming potential extrapolation issues. The developed models were also capable of predicting performance for a different CHO cell line, using similar process conditions but producing a different product. Under these conditions, the RF regressor outperformed the XGB regressor and the feed forward neural network for the prediction of VCD. Retraining the developed models on a subset of data from the second CHO cell line, further improved the predictive capability of the models. The choice of machine learning algorithm, can

therefore have an impact where there is higher variability in the unseen data, requiring an initial investigation on the most appropriate algorithm to be used based on the proposed application of the hybrid model.

CRedit authorship contribution statement

Bhagya S. Yatipanthalawa: Conceptualization, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft. **Shauna E. Wallace Fitzsimons:** Data curation, Resources, Writing – review & editing. **Tizita Horning:** Data curation, Resources. **Yih Yean Lee:** Conceptualization, Resources, Supervision, Writing – review & editing. **Sally L. Gras:** Conceptualization, Funding acquisition, Project administration, Supervision, Writing – original draft, Writing – review & editing.

Declaration of competing interest

Shauna E. Wallace Fitzsimons, Tizita Horning and Yih Yean Lee are employees of CSL Limited. Bhagya S. Yatipanthalawa and Sally L. Gras declare no competing interests, including no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Normalised data are included in a supplementary file.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.compchemeng.2024.108648](https://doi.org/10.1016/j.compchemeng.2024.108648).

References

- Badillo, S., Banfai, B., Birzele, F., Davydov, I.I., Hutchinson, L., Kam-Thong, T., Siebourg-Polster, J., Steiert, B., Zhang, J.D., 2020. An introduction to machine learning. *Clin. Pharmacol. Therapeutics* 107, 871–885.
- Bayer, B., Sissolab, B., Duerkop, M., Von Stosch, M., Striedner, G., 2020. The shortcomings of accurate rate estimations in cultivation processes and a solution for precise and robust process modeling. *Bioprocess. Biosyst. Eng.* 43, 169–178.
- Carvalho, M., Riesberg, J., Budman, H., 2022. Hybrid model to predict the effect of complex media changes in mammalian cell cultures. *Biochem. Eng. J.* 186, 108560.
- Chasiotis, V., Tzempelikos, D., Filios, A., Moustiris, K.P., 2020. Artificial neural network modelling of moisture content evolution for convective drying of cylindrical quince slices. *Comput. Electron. Agric.* 172, 105074.
- Chen, T., Guestrin, C., 2016. Xgboost: a scalable tree boosting system. In: *Proceedings of the 22nd Acm Sigkdd International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
- Del Rio-Chanona, E.A., Fiorelli, F., Zhang, D., Ahmed, N.R., Jing, K., Shah, N., 2017. An efficient model construction strategy to simulate microalgal lutein photo-production dynamic process. *Biotechnol. Bioeng.* 114, 2518–2527.
- Dong, Y., Fan, Q., Yan, X., Guo, M., Lu, F., 2014. Development of a hybrid model for sodium gluconate fermentation by *Aspergillus niger*. *J. Chem. Technol. Biotechnol.* 89, 1875–1882.
- Goldrick, S., Sandner, V., Cheeks, M., Turner, R., Farid, S.S., McCreath, G., Glassey, J., 2020. Multivariate data analysis methodology to solve data challenges related to scale-up model validation and missing data on a micro-bioreactor system. *Biotechnol. J.* 15, 1800684.
- Hassan, S.S., Farhan, M., Mangayil, R., Huttunen, H., Aho, T., 2013. Bioprocess data mining using regularized regression and random forests. *BMC Syst. Biol.* 7, 1–7.
- Hutter, C., Von Stosch, M., Cruz Bournazou, M.N., Butté, A., 2021. Knowledge transfer across cell lines using hybrid Gaussian process models with entity embedding vectors. *Biotechnol. Bioeng.* 118, 4389–4401.
- Kapoor, S., & Narayanan, A. 2022. Leakage and the reproducibility crisis in ML-based science. *arXiv preprint arXiv:2207.07048*.
- Kiangala, S.K., Wang, Z., 2021. An effective adaptive customization framework for small manufacturing plants using extreme gradient boosting-XGBoost and random forest ensemble learning algorithms in an Industry 4.0 environment. *Mach. Learn. Appl.* 4, 100024.
- Klein, T., Heinzel, N., Kroll, P., Brunner, M., Herwig, C., Neutsh, L., 2015. Quantification of cell lysis during CHO bioprocesses: impact on cell count, growth kinetics and productivity. *J. Biotechnol.* 207, 67–76.
- Lagergren, J.H., Nardini, J.T., Baker, R.E., Simpson, M.J., Flores, K.B., 2020. Biologically-informed neural networks guide mechanistic modeling from sparse experimental data. *PLoS Comput. Biol.* 16, e1008462.
- Lancashire, L.J., Lemetre, C., Ball, G.R., 2009. An introduction to artificial neural networks in bioinformatics—Application to complex microarray and mass spectrometry datasets in cancer studies. *Brief. Bioinformatics* 10, 315–329.
- Laursen, S.O., Webb, D., Ramirez, W.F., 2007. Dynamic hybrid neural network model of an industrial fed-batch fermentation process to produce foreign protein. *Comput. Chem. Eng.* 31, 163–170.
- Luna, M.F., Ochsner, A.M., Amstutz, V., Von Blarer, D., Sokolov, M., Arosio, P., Zinn, M., 2021. Modeling of continuous PHA production by a hybrid approach based on first principles and machine learning. *Processes* 9, 1560.
- Melcher, M., Scharl, T., Spangl, B., Luchner, M., Cserjan, M., Bayer, K., Leisch, F., Striedner, G., 2015. The potential of random forest and neural networks for biomass and recombinant protein modeling in *Escherichia coli* fed-batch fermentations. *Biotechnol. J.* 10, 1770–1782.
- Narayanan, H., Sokolov, M., Butté, A., Morbidelli, M., 2019a. Decision Tree-PLS (DT-PLS) algorithm for the development of process: specific local prediction models. *Biotechnol. Prog.* 35, e2818.
- Narayanan, H., Sokolov, M., Morbidelli, M., Butté, A., 2019b. A new generation of predictive models: the added value of hybrid models for manufacturing processes of therapeutic proteins. *Biotechnol. Bioeng.* 116, 2540–2549.
- Nikita, S., Thakur, G., Jesubalan, N.G., Kulkarni, A., Yezhuvath, V.B., Rathore, A.S., 2022. AI-ML applications in bioprocessing: ML as an enabler of real time quality prediction in continuous manufacturing of mAbs. *Comput. Chem. Eng.* 164, 107896.
- Oliveira, R., 2004. Combining first principles modelling and artificial neural networks: a general framework. *Comput. Chem. Eng.* 28, 755–766.
- Pardo-Diaz, J., Bozhilova, L.V., Beguerisse-Diaz, M., Poole, P.S., Deane, C.M., Reinert, G., 2021. Robust gene coexpression networks using signed distance correlation. *Bioinformatics* 37, 1982–1989.
- Park, S.Y., Kim, S.J., Park, C.H., Kim, J., Lee, D.Y., 2023. Data-driven prediction models for forecasting multistep ahead profiles of mammalian cell culture toward bioprocess digital twins. *Biotechnol. Bioeng.*
- Pinto, J., Mestre, M., Ramos, J., Costa, R.S., Striedner, G., Oliveira, R., 2022. A general deep hybrid model for bioreactor systems: combining first principles with deep neural networks. *Comput. Chem. Eng.* 165, 107952.
- Roubos, J., Krabben, P., Setnes, M., Babuska, R., Heijnen, J., Verbruggen, H., 1999. Hybrid Model Development For Fed-Batch Bioprocesses; Combining Physical Equations With the Metabolic Network and Black-Box kinetics. 6th Workshop on Fuzzy Systems. Citeseer, pp. 8–9.
- Sauer, P.W., Burky, J.E., Wesson, M.C., Sternard, H.D., Qu, L., 2000. A high-yielding, generic fed-batch cell culture process for production of recombinant antibodies. *Biotechnol. Bioeng.* 67, 585–597.
- Schubert, J., Simutis, R., Dors, M., Havlik, I., Lübbert, A., 1994. Bioprocess optimization and control: application of hybrid modelling. *J. Biotechnol.* 35, 51–68.
- Smiatek, J., Clemens, C., Herrera, L.M., Arnold, S., Knapp, B., Presser, B., Jung, A., Wucherpfennig, T., Bluhmki, E., 2021. Generic and specific recurrent neural network models: applications for large and small scale biopharmaceutical upstream processes. *Biotechnol. Rep.* 31, e00640.
- Sokolov, M., Ritscher, J., Mackinnon, N., Souquet, J., Broly, H., Morbidelli, M., Butté, A., 2017. Enhanced process understanding and multivariate prediction of the relationship between cell culture process and monoclonal antibody quality. *Biotechnol. Rep.* 33, 1368–1380.
- Stefas, D., Gyftokostas, N., Nanou, E., Kourelia, P., Couris, S., 2021. Laser-Induced Breakdown Spectroscopy: an efficient tool for food science and technology (from the analysis of martian rocks to the analysis of olive oil, honey, milk, and other natural earth products). *Molecules* 26, 4981.
- Teixeira, A.P., Alves, C., Alves, P.M., Carrondo, M.J., Oliveira, R., 2007. Hybrid elementary flux analysis/nonparametric modeling: application for bioprocess control. *BMC Bioinformatics* 8, 1–15.
- Von Stosch, M., Hamelink, J.M., Oliveira, R., 2016. Hybrid modeling as a QbD/PAT tool in process development: an industrial *E. coli* case study. *Bioprocess. Biosyst. Eng.* 39, 773–784.
- Xu, P., Ji, X., Li, M., Lu, W., 2023. Small data machine learning in materials science. *NPJ Comput. Mater.* 9, 42.
- Yang, A., Martin, E., Morris, J., 2011. Identification of semi-parametric hybrid process models. *Comput. Chem. Eng.* 35, 63–70.